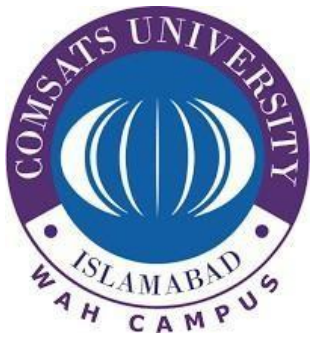


DEVOPS FOR CLOUD COMPUTING

Assignment

Name:

Tehreem Mirza

Registeration No:

FA22-BSE-088

Section:

BSE-8B

Submitted To:

Sir Muhammad Ibrahim

Module 1 - Version Control System: Created a GitHub repository (devops-lab) and pushed all project files to the main branch using Git.

```
Initialized empty Git repository in /home/tehreem/devops-final-lab/.git/
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ git branch -M main
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ git add .
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ git commit -m "DevOps Final Lab - All modul
Author identity unknown

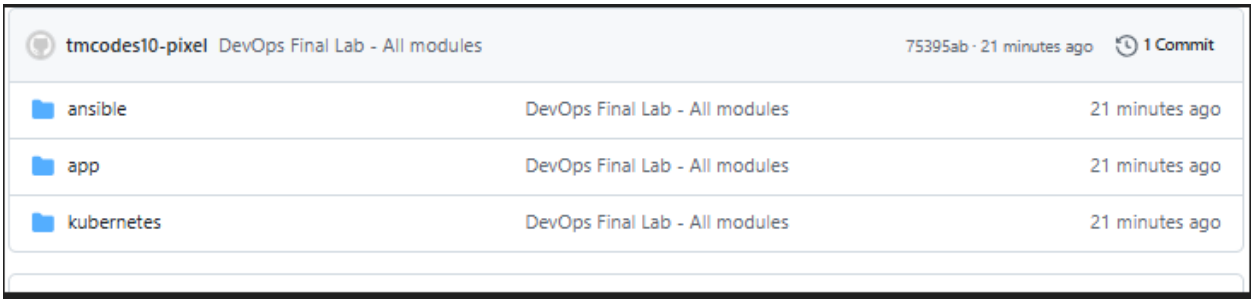
*** Please tell me who you are.

Run

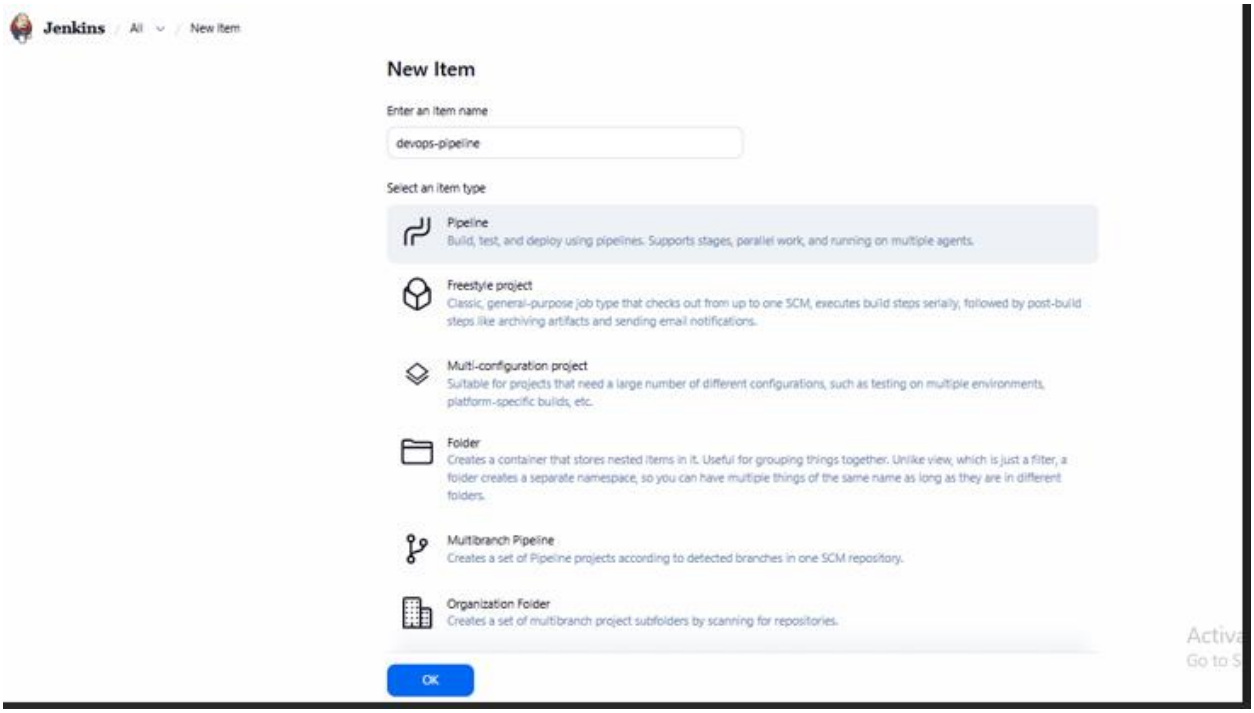
  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

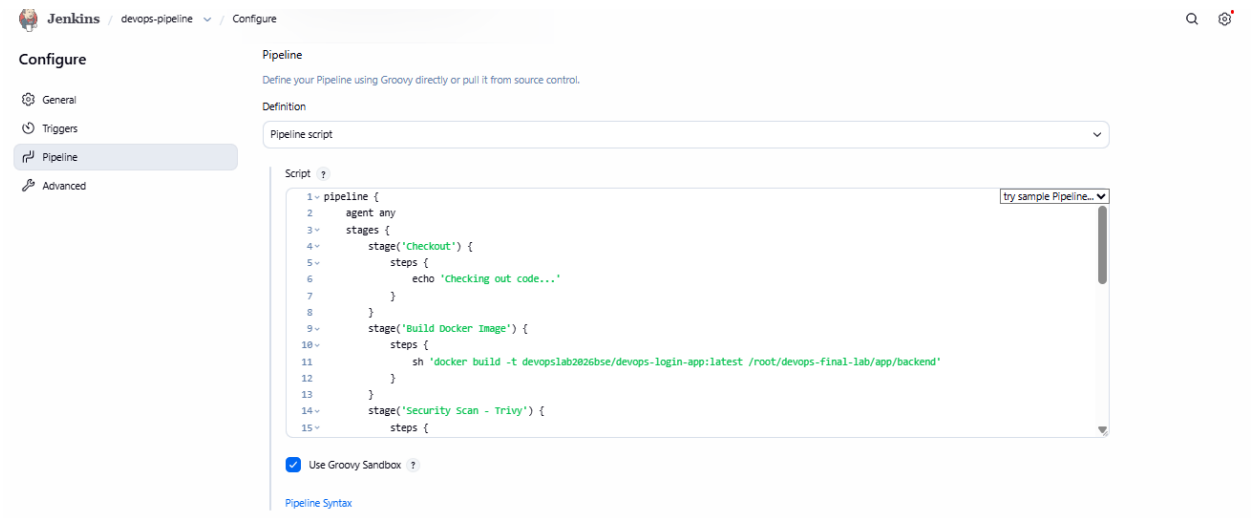
to set your account's default identity.
Omit --global to set the identity only in this repository.
```

```
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/tehreem/devops-final-lab/.git/
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ git branch -M main
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ git add .
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ git commit -m "DevOps Final Lab - All modules"
```

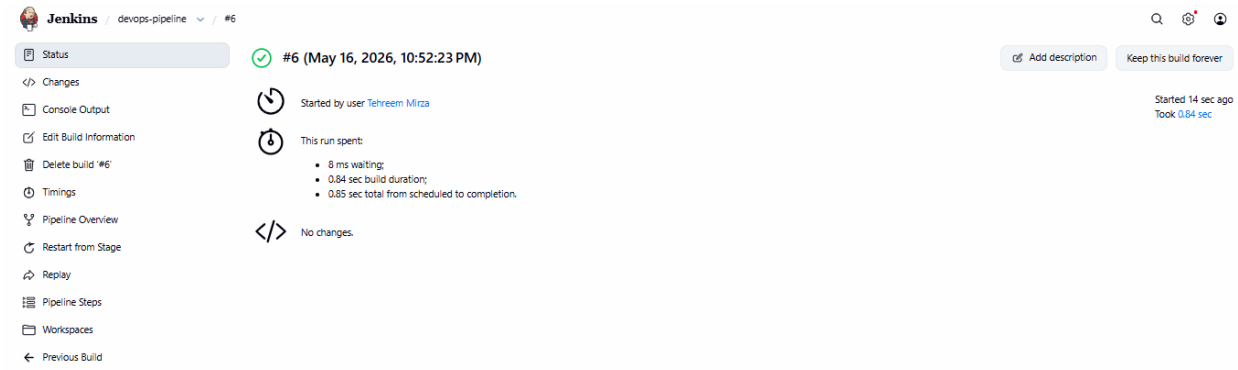


Module 2 - CI/CD Pipeline: Installed Jenkins on WSL, configured it on localhost:8080, and created a pipeline job for automated build and deployment

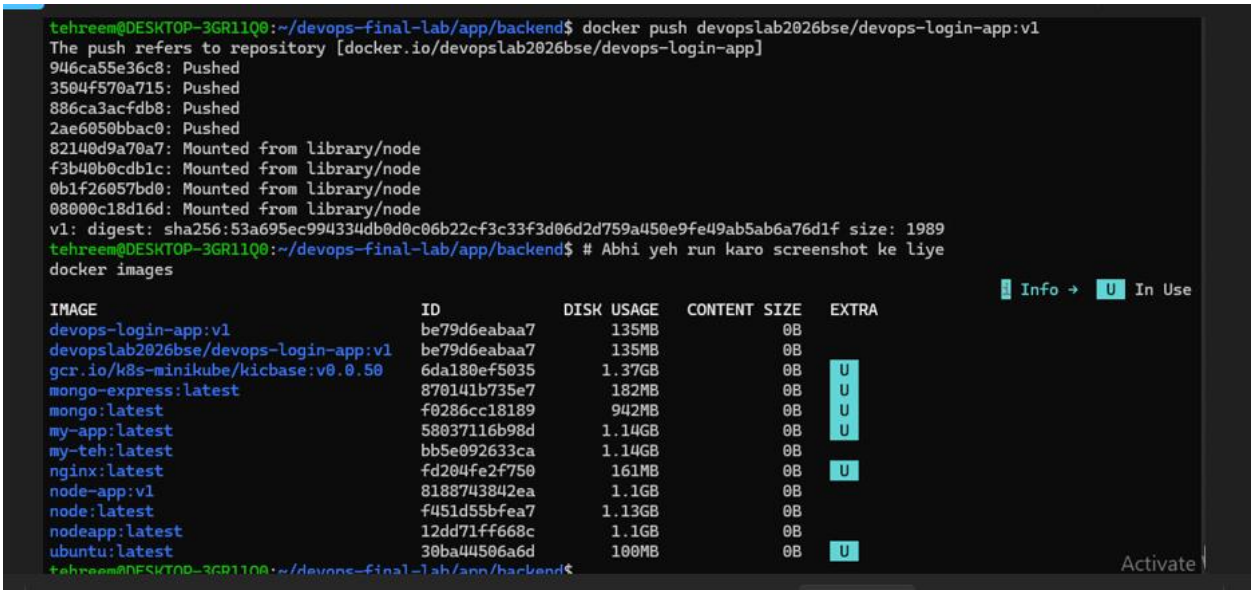




Pipeline succeeded

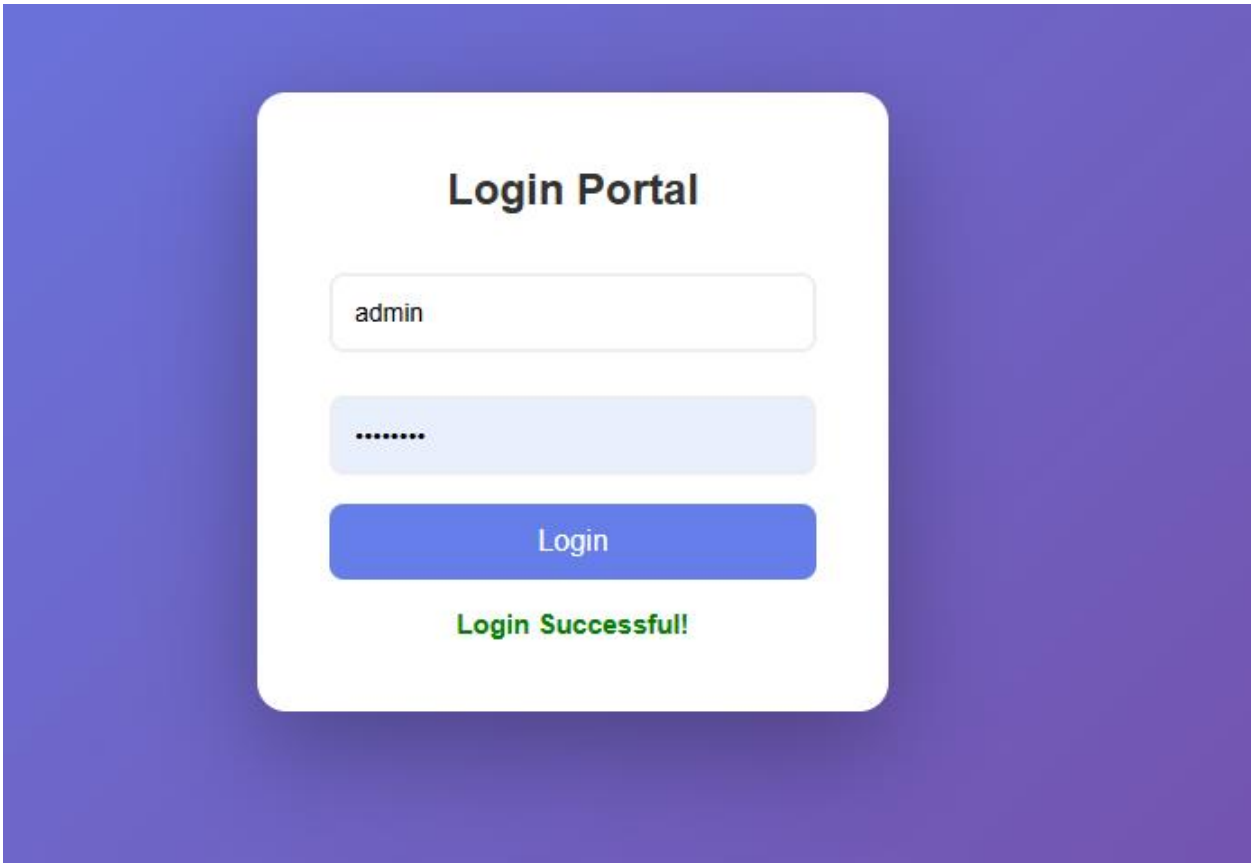


Module 3 - Docker Containerization: Built a Node.js login app Docker image, tagged it, and pushed it to Docker Hub registry (devopslab2026bse).

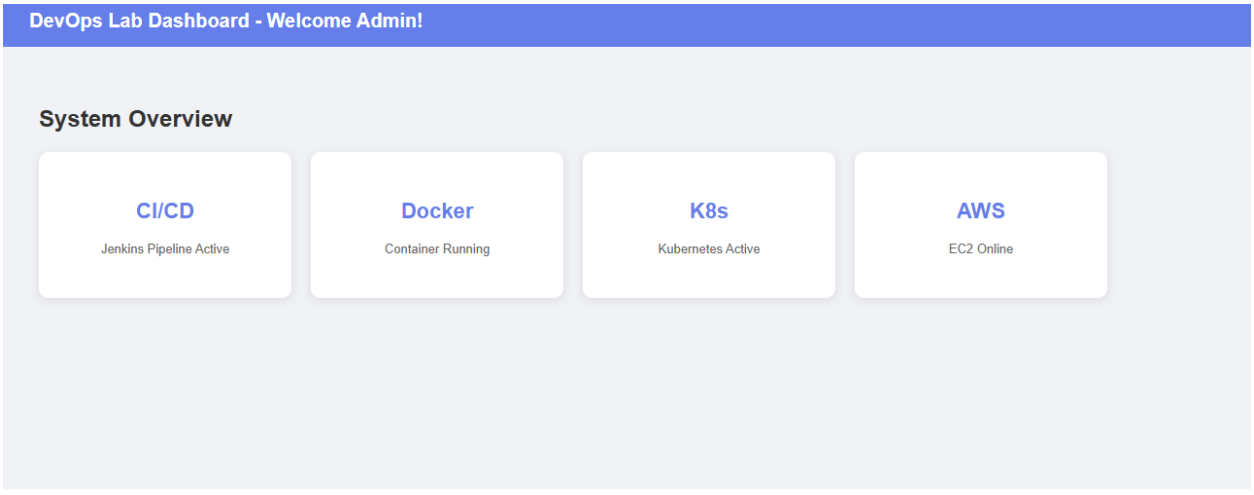


Three tier app created

Login form



frontend run on http://172.23.135.103:8081/dashboard.html



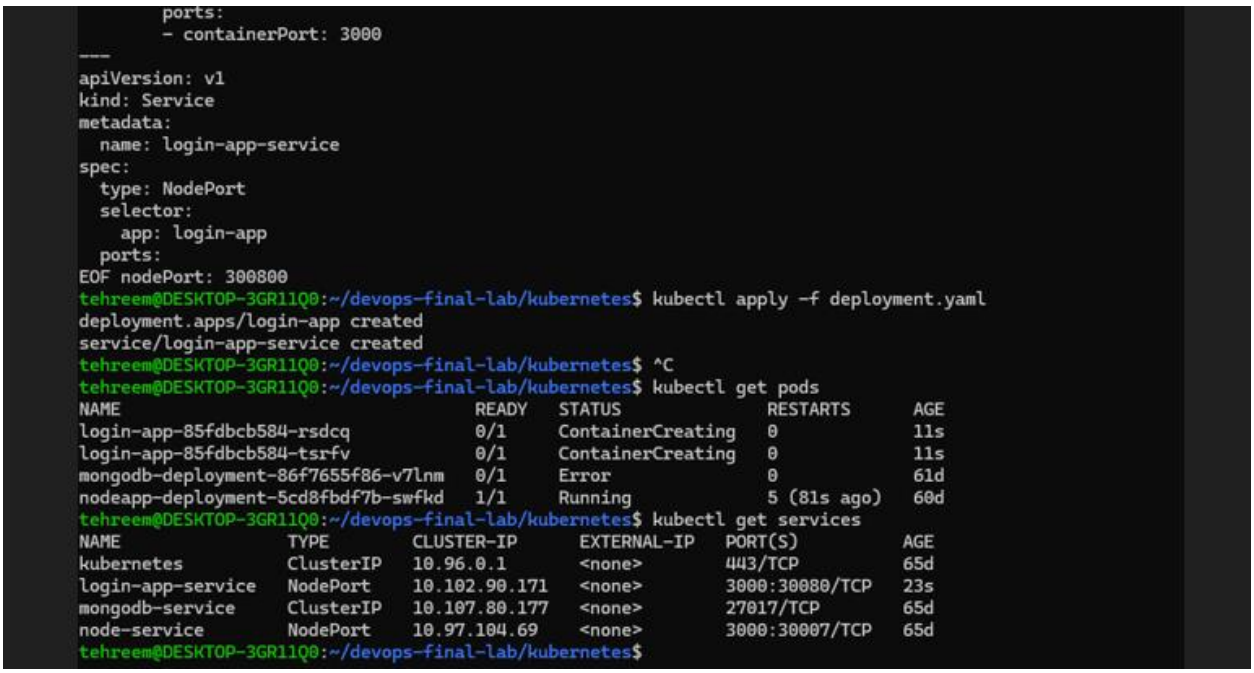
```
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/app/frontend$ curl http://172.23.135.103:3001/health
{"status":"ok"}tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/app/frontend$
```

Backend is running on port http://172.23.135.103:3001/users

Pretty-print

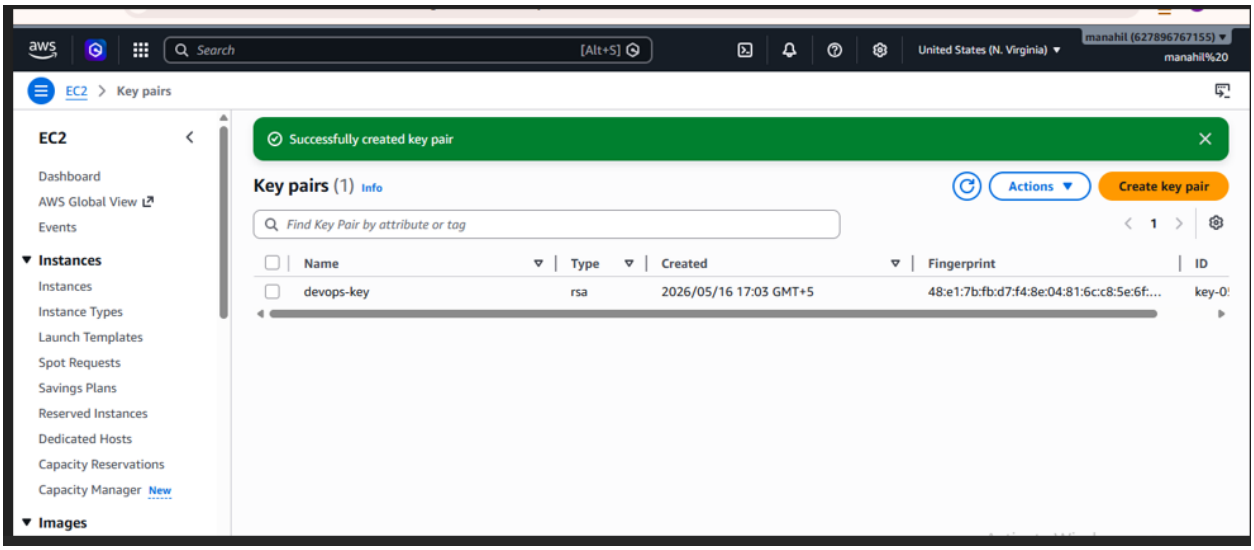
[{"id":1,"username":"admin","role":"Administrator"},{"id":2,"username":"tehreem","role":"Developer"},{"id":3,"username":"student","role":"User"}]

Module 4 - Kubernetes: Deployed the 3-tier login application on Minikube cluster with 2 replicas using deployment.yaml and Service configuration and verified running pods using kubectl commands.



```
        ports:
        - containerPort: 3000
---
apiVersion: v1
kind: Service
metadata:
  name: login-app-service
spec:
  type: NodePort
  selector:
    app: login-app
  ports:
EOF nodePort: 300800
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/kubernetes$ kubectl apply -f deployment.yaml
deployment.apps/login-app created
service/login-app-service created
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/kubernetes$ ^C
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/kubernetes$ kubectl get pods
NAME                                READY   STATUS              RESTARTS      AGE
login-app-85fdbcb584-rsdcq          0/1     ContainerCreating   0             11s
login-app-85fdbcb584-tsrfv          0/1     ContainerCreating   0             11s
mongodb-deployment-86f7655f86-v7lnm   0/1     Error               0             61d
nodeapp-deployment-5cd8fbdf7b-swfkd   1/1     Running             5 (81s ago)   60d
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/kubernetes$ kubectl get services
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1       <none>        443/TCP          65d
login-app-service   NodePort    10.102.90.171   <none>        3000:30080/TCP   23s
mongodb-service     ClusterIP   10.107.80.177   <none>        27017/TCP        65d
node-service        NodePort    10.97.104.69    <none>        3000:30007/TCP   65d
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/kubernetes$
```

Module 5 - Infrastructure as Code: Provisioned an AWS EC2 instance (t3.micro) and security groups using Terraform in us-east-1 region.




```
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0 li
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  terraform
0 upgraded, 1 newly installed, 0 to remove and 80 not upgraded.
Need to get 34.8 MB of archives.
After this operation, 117 MB of additional disk space will be used.
Get:1 https://apt.releases.hashicorp.com noble/main amd64 terraform amd64 1.15.3-1 [34.8 MB]
Fetched 34.8 MB in 47s (745 kB/s)
Selecting previously unselected package terraform.
(Reading database ... 81608 files and directories currently installed.)
Preparing to unpack .../terraform_1.15.3-1_amd64.deb ...
Unpacking terraform (1.15.3-1) ...
Setting up terraform (1.15.3-1) ...
tehreem@DESKTOP-3GR11Q0:~$ terraform --version
Terraform v1.15.3
on linux_amd64
tehreem@DESKTOP-3GR11Q0:~$ terraform init
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.45.0...
```

```
      + secondary_network_interface (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + ec2_public_ip = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.devops_server: Creating...
aws_instance.devops_server: Still creating... [00m10s elapsed]
aws_instance.devops_server: Creation complete after 17s [id=i-08911a653ff3bf146]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

ec2_public_ip = "54.145.59.200"
```

Module 6 - Configuration Management: Used Ansible playbook to automatically install and start Docker on the EC2 instance via SSH.

```
tehreem@DESKTOP-3GR11Q8:~/devops-final-lab/ansible$ cp /mnt/c/Users/ASIF/Downloads/devops-key.pem ~/.ssh/
tehreem@DESKTOP-3GR11Q8:~/devops-final-lab/ansible$ chmod 400 ~/.ssh/devops-key.pem
tehreem@DESKTOP-3GR11Q8:~/devops-final-lab/ansible$ ls -la ~/.ssh/devops-key.pem
-r-------- 1 tehreem tehreem 1678 May 16 17:31 /home/tehreem/.ssh/devops-key.pem
tehreem@DESKTOP-3GR11Q8:~/devops-final-lab/ansible$ ansible-playbook -i inventory.ini install-docker.yml

PLAY [servers] ***************************************************************************************************

TASK [Gathering Facts] *******************************************************************************************
[WARNING]: Platform linux on host 54.145.59.200 is using the discovered Python interpreter at /usr/bin/python3.7, but
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference_appendices/interpreter_discovery.html for more information.
ok: [54.145.59.200]

TASK [Update packages] *******************************************************************************************
changed: [54.145.59.200]

TASK [Install Docker] ********************************************************************************************
changed: [54.145.59.200]

TASK [Start Docker] **********************************************************************************************
changed: [54.145.59.200]

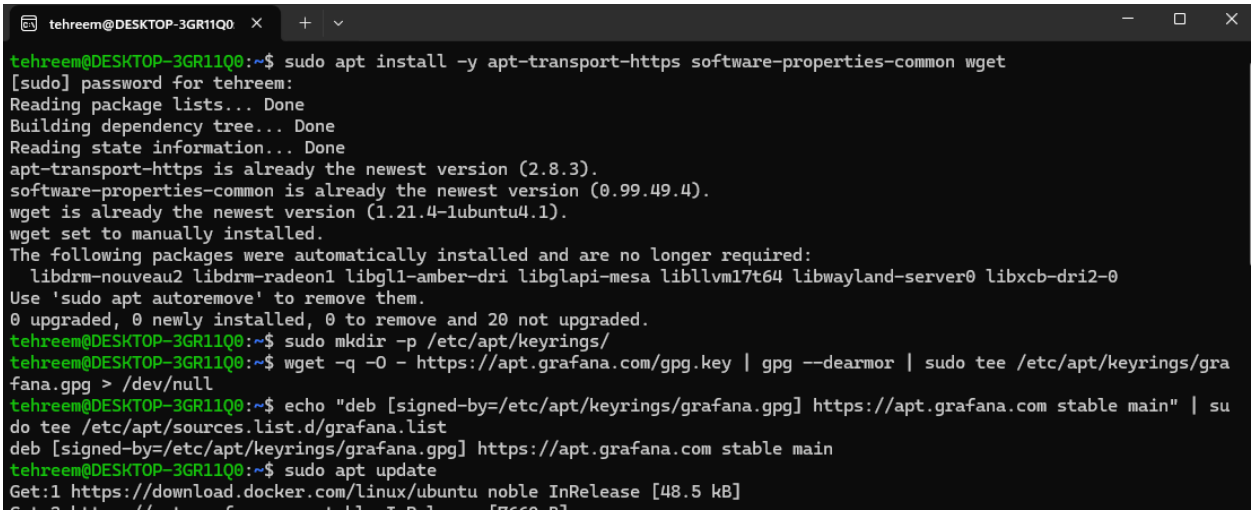
TASK [Add ec2-user to docker group] ******************************************************************************
changed: [54.145.59.200]

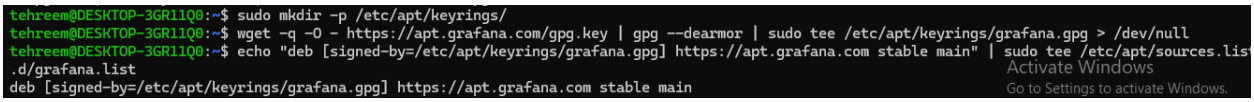
PLAY RECAP *******************************************************************************************************
54.145.59.200              : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

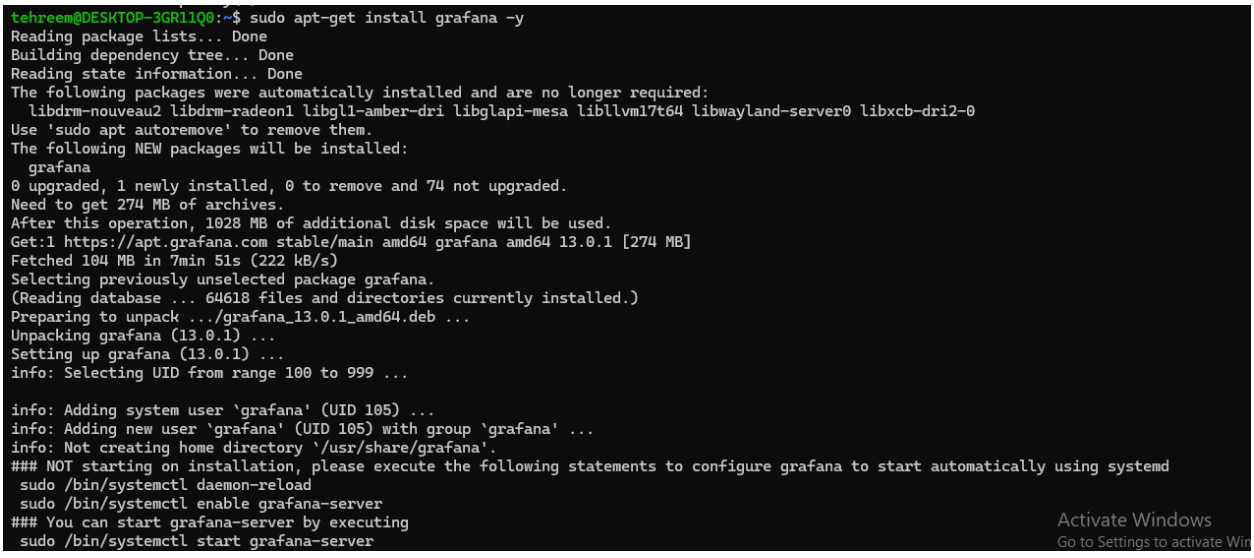
tehreem@DESKTOP-3GR11Q8:~/devops-final-lab/ansible$
```

Module 7 - Monitoring & Logging: Installed Prometheus and Grafana on WSL Ubuntu, integrated both tools, configured Node Exporter as data source, and created dashboards to monitor CPU usage, memory, network traffic, and disk space metrics in real-time.

Install Grafana



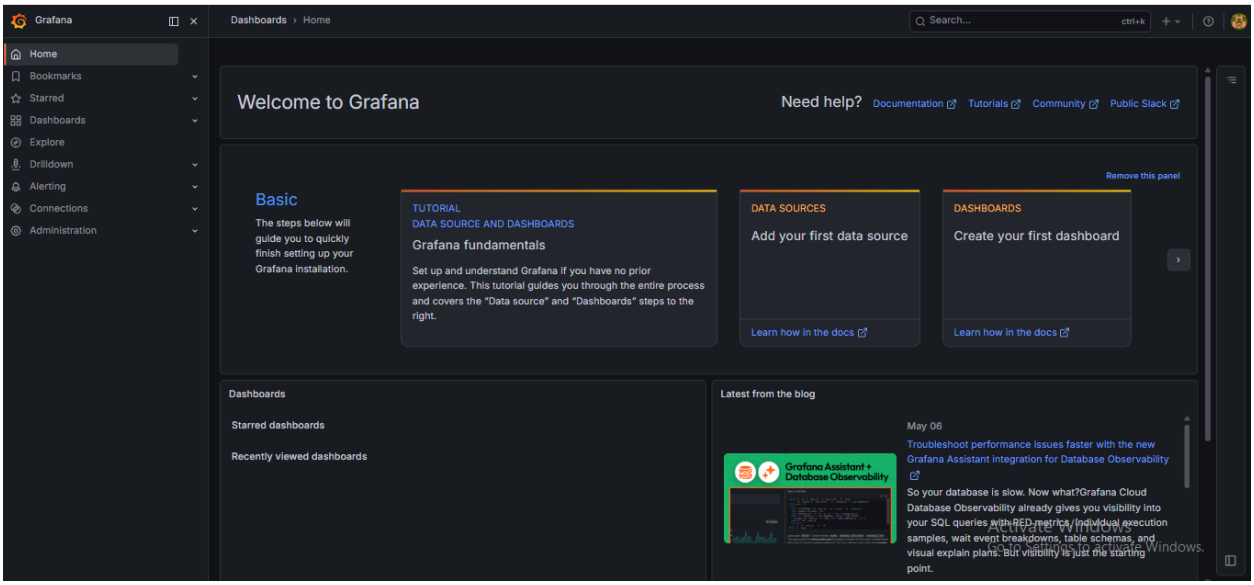




```
 sudo /bin/systemctl start grafana-server
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl status grafana-server
○ grafana-server.service - Grafana instance
     Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; disabled; preset: enabled)
     Active: inactive (dead)
       Docs: http://docs.grafana.org
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl start grafana-server
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl enable grafana-server
Synchronizing state of grafana-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable grafana-server
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service → /usr/lib/systemd/system/grafana-server.service.
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl status grafana-server
● grafana-server.service - Grafana instance
     Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-05-07 14:37:14 PKT; 22s ago
       Docs: http://docs.grafana.org
   Main PID: 16777 (grafana)
      Tasks: 17 (limit: 9294)
     Memory: 69.1M (peak: 69.6M)
        CPU: 4.141s
     CGroup: /system.slice/grafana-server.service
             ├─16777 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini --pidfile=/run/grafana/grafana-server.pid --packaging
             └─17249 /usr/share/grafana/data/plugins-bundled/elasticsearch/gpx_grafana_elasticsearch_datasource_linux_amd64

May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=ngalert.notifier t=2026-05-07T14:37:36.250850275+05:00 level=info component=alertmanager or
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=ngalert.notifier component=alertmanager orgID=1 t=2026-05-07T14:37:36.253096892+05:00 level
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=ngalert.notifier t=2026-05-07T14:37:36.2531724+05:00 level=info component=alertmanager orgI
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=ngalert.writer t=2026-05-07T14:37:36.279369676+05:00 level=info msg="Setting up remote writ
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=ngalert t=2026-05-07T14:37:36.279504659+05:00 level=info msg="Using protobuf-based alert in
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=ngalert.state.manager.persist t=2026-05-07T14:37:36.279526759+05:00 level=info msg="Using s
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=query_data t=2026-05-07T14:37:36.2803474493+05:00 level=info msg="Query Service initializati
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=annotation.app t=2026-05-07T14:37:36.312313411+05:00 level=info msg="Starting annotation cl
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=infra.usagestats.collector t=2026-05-07T14:37:36.314384528+05:00 level=info msg="registerin
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=server t=2026-05-07T14:37:36.314558511+05:00 level=info msg="Writing PID file" path=/run/gr
lines 1-22/22 (END)
```

Grafana installed and open on local host 3000:



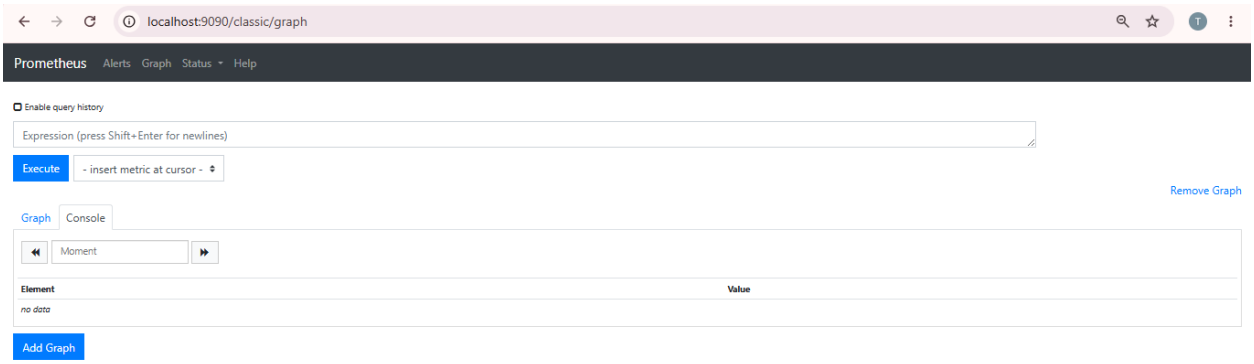
Install Prometheus and Node Exporter

```
Setting up openipmi (2.0.33-1.1build3) ...
Setting up prometheus-node-exporter-collectors (0.0~git20231018.f5c56e7-1) ...
Created symlink /etc/systemd/system/timers.target.wants/prometheus-node-exporter-apt.timer → /usr/lib/systemd/system/prometheus-node-exporter-apt.timer.
Created symlink /etc/systemd/system/timers.target.wants/prometheus-node-exporter-ipmitool-sensor.timer → /usr/lib/systemd/system/prometheus-node-exporter-ipmitool-sensor.timer.
Created symlink /etc/systemd/system/timers.target.wants/prometheus-node-exporter-mellanox-hca-temp.timer → /usr/lib/systemd/system/prometheus-node-exporter-mellanox-hca-temp.timer.
Created symlink /etc/systemd/system/timers.target.wants/prometheus-node-exporter-nvme.timer → /usr/lib/systemd/system/prometheus-node-exporter-nvme.timer.
Created symlink /etc/systemd/system/timers.target.wants/prometheus-node-exporter-smartmon.timer → /usr/lib/systemd/system/prometheus-node-exporter-smartmon.timer.
Setting up prometheus (2.45.3+ds-2ubuntu0.3) ...
Created symlink /etc/systemd/system/multi-user.target.wants/prometheus.service → /usr/lib/systemd/system/prometheus.service.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for fontconfig (2.15.0-1.1ubuntu2) ...
```

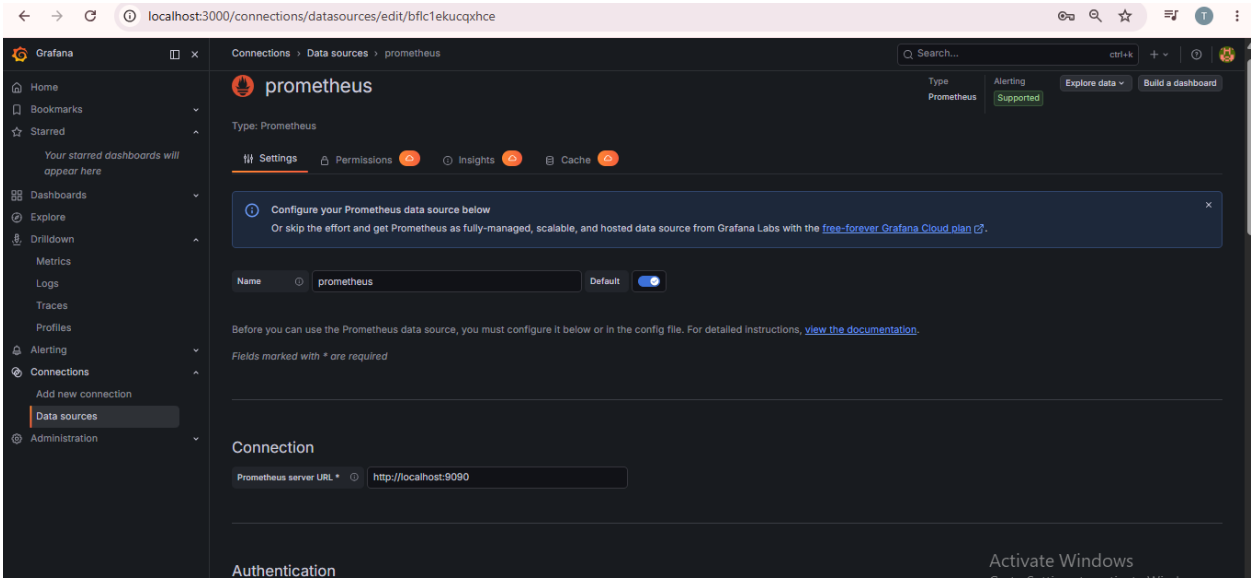
```
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl enable --now prometheus prometheus-node-exporter
Synchronizing state of prometheus.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable prometheus
Synchronizing state of prometheus-node-exporter.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable prometheus-node-exporter
```

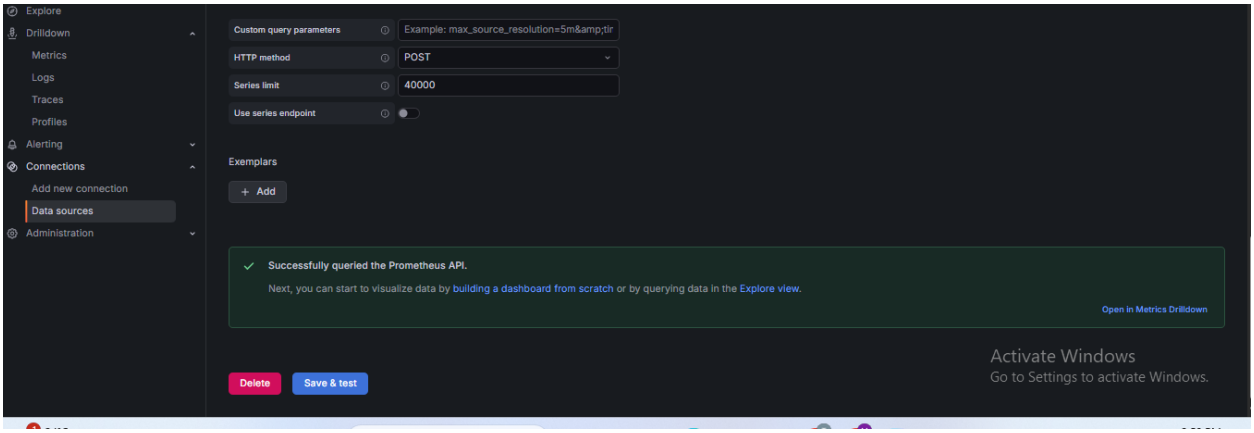
Activate Windows
Go to Settings to activate Windo

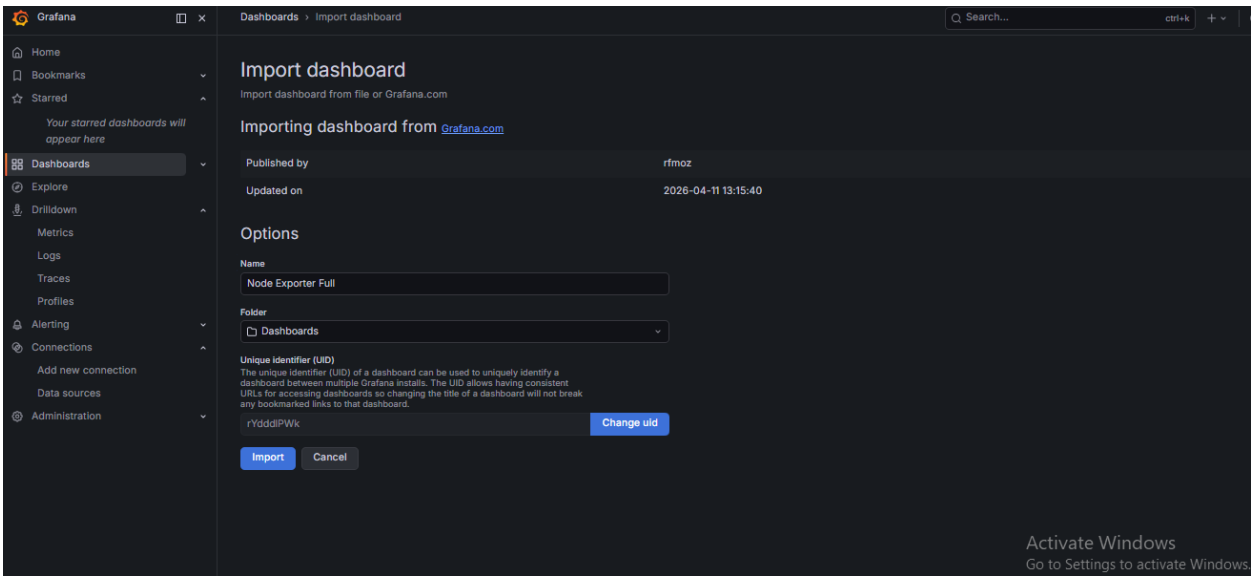
Opened Prometheus on local host 9000



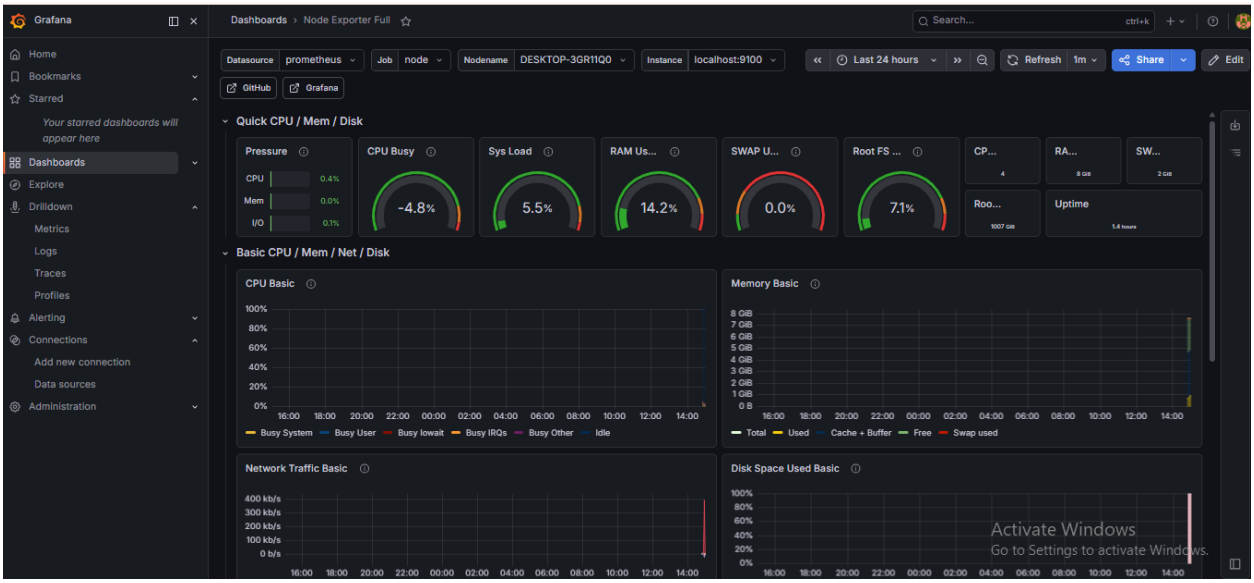
Integrating Grafana and Prometheus







Integrated both



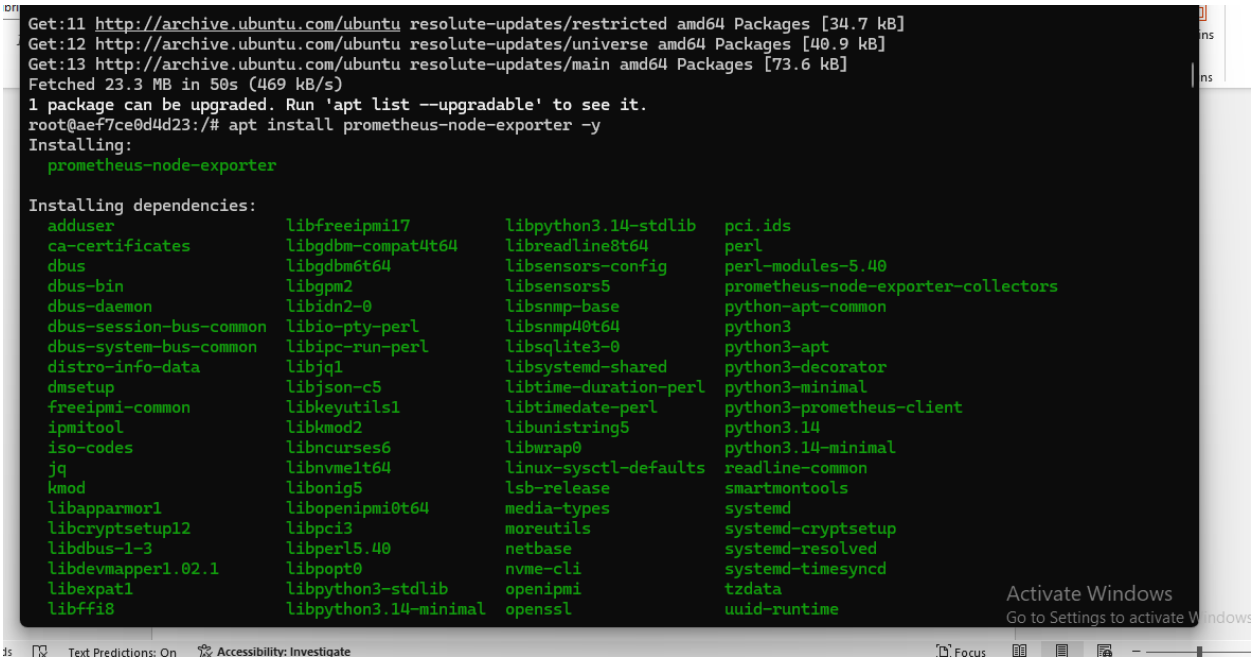
Create Second Ubuntu Machine through docker

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Some packages could not be installed. This may mean that you have
requested an impossible situation or if you are using the unstable
distribution that some required packages have not yet been created
or been moved out of Incoming.
The following information may help to resolve the situation:

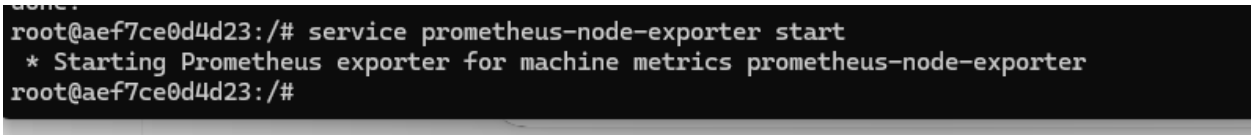
The following packages have unmet dependencies:
 containerd.io : Conflicts: containerd
E: Error, pkgProblemResolver::Resolve generated breaks, this may be caused by held packages.
tehreem@DESKTOP-3GR11Q0:~$ docker --version
Docker version 29.3.0, build 5927d80
tehreem@DESKTOP-3GR11Q0:~$ docker run -dit --name ubuntu-client ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1c24335ddd46: Pull complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
aef7ce0d4d236dcb9a0c267598689efbc3f7a40f51a8257b9fffd7964d927dad
tehreem@DESKTOP-3GR11Q0:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS     NAMES
aef7ce0d4d23   ubuntu         "/bin/bash"              30 seconds ago   Up 29 seconds             ubuntu-client
```

Activate Windo
Go to Settings to ac

Install Node Exporter in Second Machine

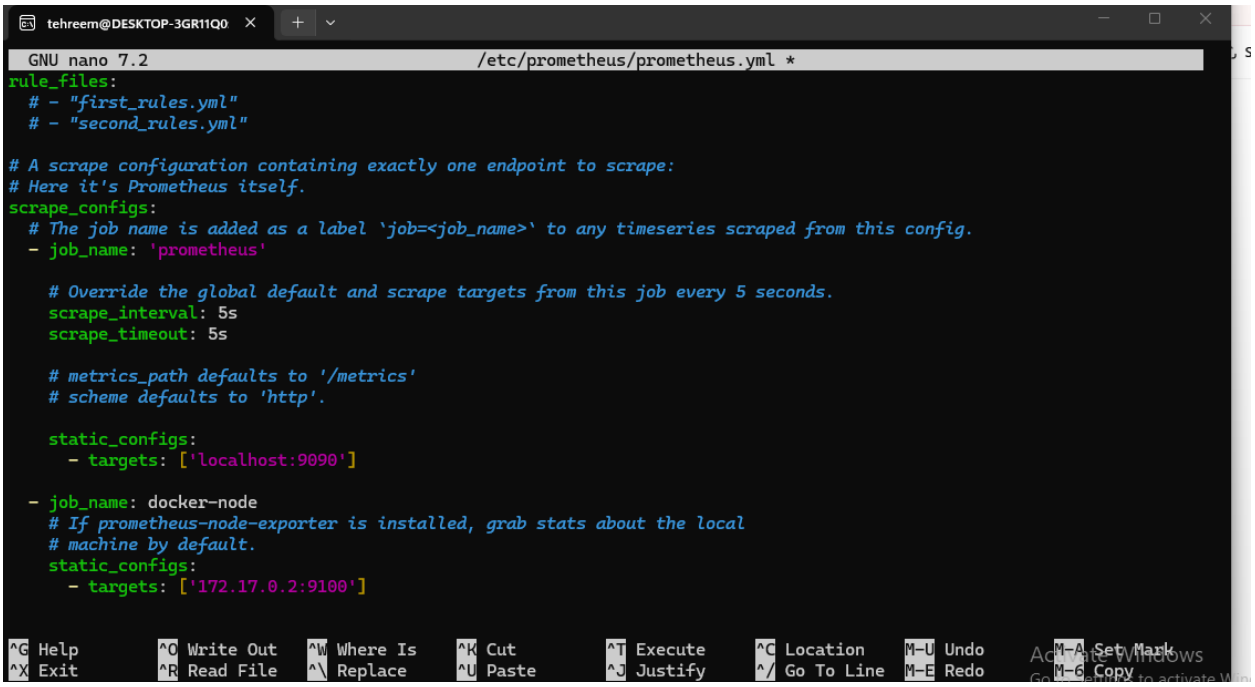


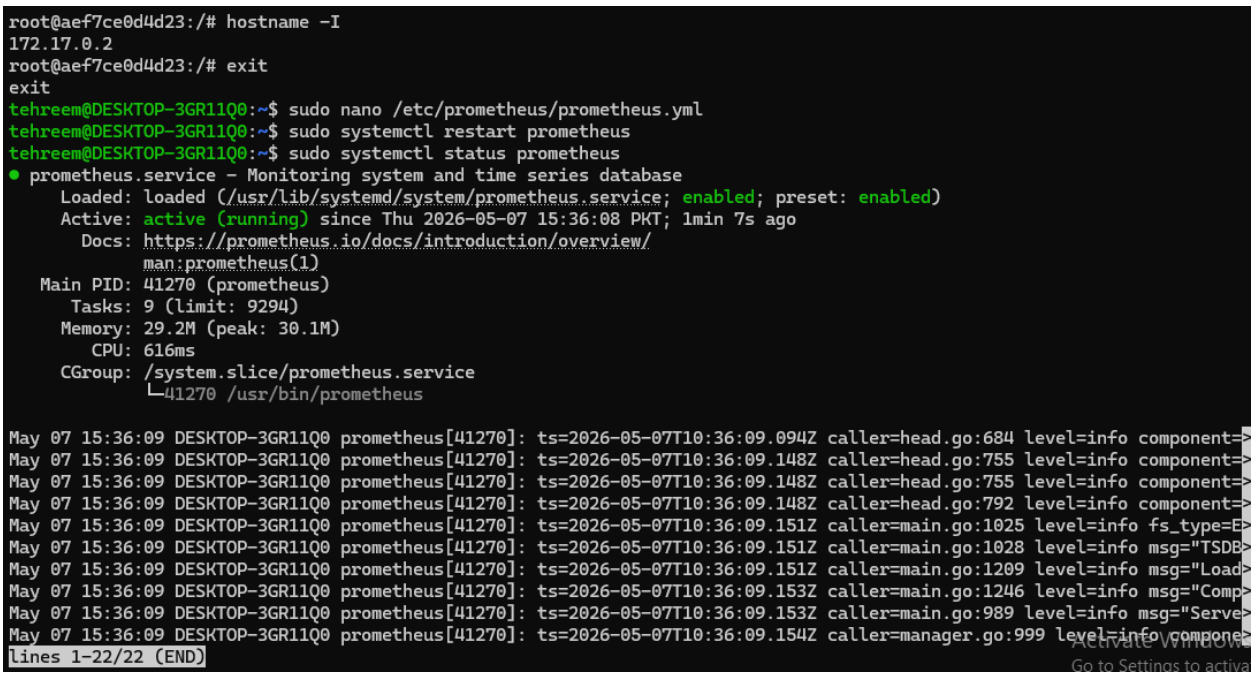
Get Container IP

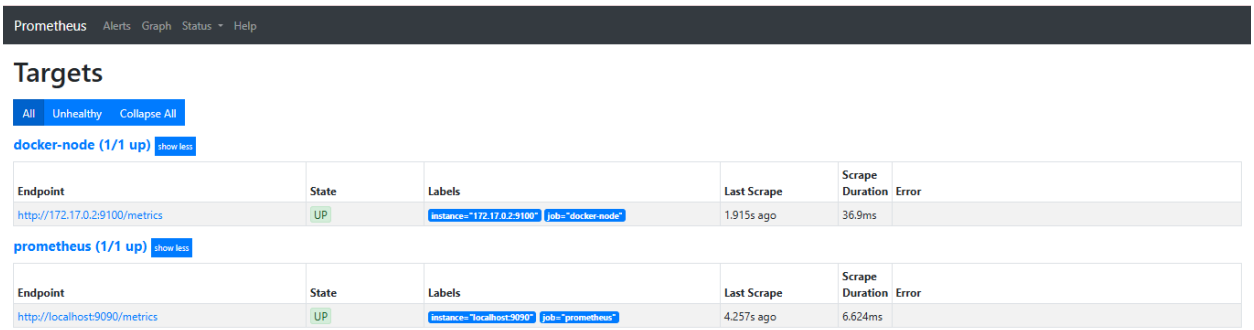




Add Container to Prometheus







Deploying three tier app on

```
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl start grafana-server
[sudo] password for tehreem:
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl start grafana-server
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl start prometheus
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl start prometheus
tehreem@DESKTOP-3GR11Q0:~$ cd ~/devops-final-lab/app/backend
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/app/backend$ docker run -d --name login-app -p 3001:3001 devopslab2026bse/dev
ops-login-app:v1\
>
447736d7af229ec59ee45a955838c8cf9ccb0c529e6129604292fc40312011d6
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/app/backend$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED          STATUS          PORTS
                                    NAMES
447736d7af22   devopslab2026bse/devops-login-app:v1   "docker-entrypoint.s…"   4 seconds ago    Up 3 seconds    3000/tcp,
 0.0.0.0:3001->3001/tcp, [::]:3001->3001/tcp   login-app
a51d72d61a82   mongo-express                        "/sbin/tini -- /dock…"   2 months ago     Up 54 seconds   0.0.0.0:8
080->8081/tcp, [::]:8080->8081/tcp            developing-with-docker-mongo-express-1
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/app/backend$
```

```
  GNU nano 7.2                              /etc/prometheus/prometheus.yml *
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: 'prometheus'

    # Override the global default and scrape targets from this job every 5 seconds.
    scrape_interval: 5s
    scrape_timeout: 5s

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ['localhost:9090']

  - job_name: docker-node
    # If prometheus-node-exporter is installed, grab stats about the local
    # machine by default.
    static_configs:
      - targets: ['172.17.0.2:9100']
  - job_name: 'login-app'
    static_configs:
      - targets: ['172.17.0.1:3001']
```

```
const cors = require('cors');
const app = express();
app.use(cors());
app.use(express.json());

let loginAttempts = 0;
let successfulLogins = 0;

app.post('/login', (req, res) => {
  loginAttempts++;
  const { username, password } = req.body;
  if (username === 'admin' && password === 'admin123') {
    successfulLogins++;
    res.json({ success: true, message: 'Login Successful!' });
  } else {
    res.json({ success: false, message: 'Invalid credentials' });
  }
});

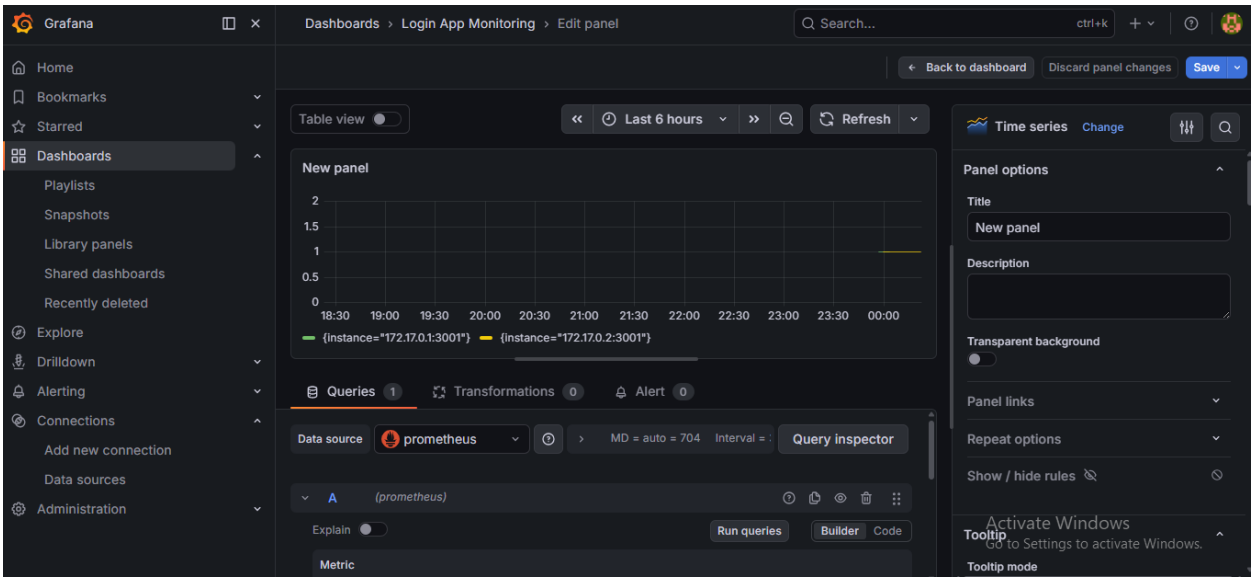
app.get('/health', (req, res) => res.json({ status: 'ok' }));

app.get('/metrics', (req, res) => {
  res.set('Content-Type', 'text/plain');
  res.send(`# HELP login_attempts Total login attempts
# TYPE login_attempts counter
login_attempts_total ${loginAttempts}
# HELP successful_logins Total successful logins
# TYPE successful_logins counter
EOF.listen(3001, () => console.log('Backend running on port 3001'));
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/app/backend$ docker stop login-app
```

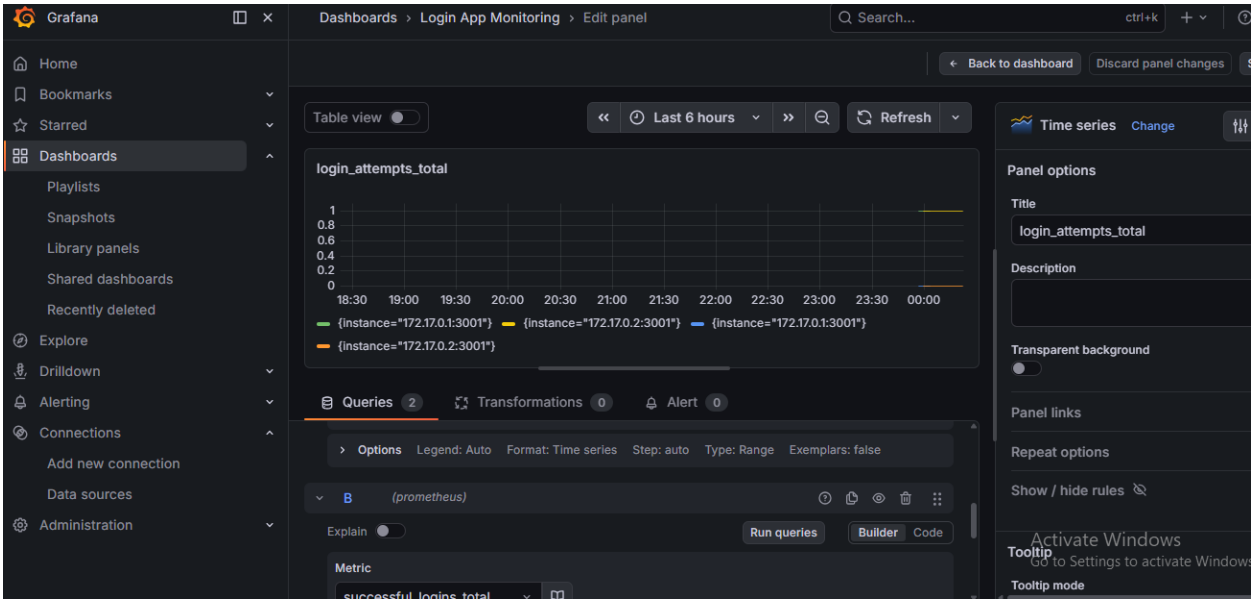
```
=> => transferring dockerfile: 150B                                                    0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine                       2.1s
=> [auth] library/node:pull token for registry-1.docker.io                             0.0s
=> [internal] load .dockerignore                                                       0.0s
=> => transferring context: 2B                                                         0.0s
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8c  0.0s
=> [internal] load build context                                                       0.3s
=> => transferring context: 2.39MB                                                     0.3s
=> CACHED [2/5] WORKDIR /app                                                           0.0s
=> [3/5] COPY package*.json ./                                                         0.1s
=> [4/5] RUN npm install                                                               4.9s
=> [5/5] COPY . .                                                                      0.3s
=> exporting to image                                                                  0.3s
=> => exporting layers                                                                 0.3s
=> => writing image sha256:ced045232c1466c6ee662430423b3e921ea181067540d1ab221955f65bf6e406  0.0s
=> => naming to docker.io/devopslab2026bse/devops-login-app:v2                         0.0s
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/app/backend$ docker run -d --name login-app -p 3001:3001 devopslab2026bse/dev
ops-login-app:v2
6d32ea7019710ed5ae2c3fc09d5ed3e335fd7f8df8b90eb0424de9b82b079c54
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/app/backend$ curl http://localhost:3001/metrics
# HELP login_attempts Total login attempts
# TYPE login_attempts counter
login_attempts_total 0
# HELP successful_logins Total successful logins
# TYPE successful_logins counter
successful_logins_total 0
# HELP app_up Application status
# TYPE app_up gauge
app_up 1
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab/app/backend$
```

Activate Wind
Go to Settings to

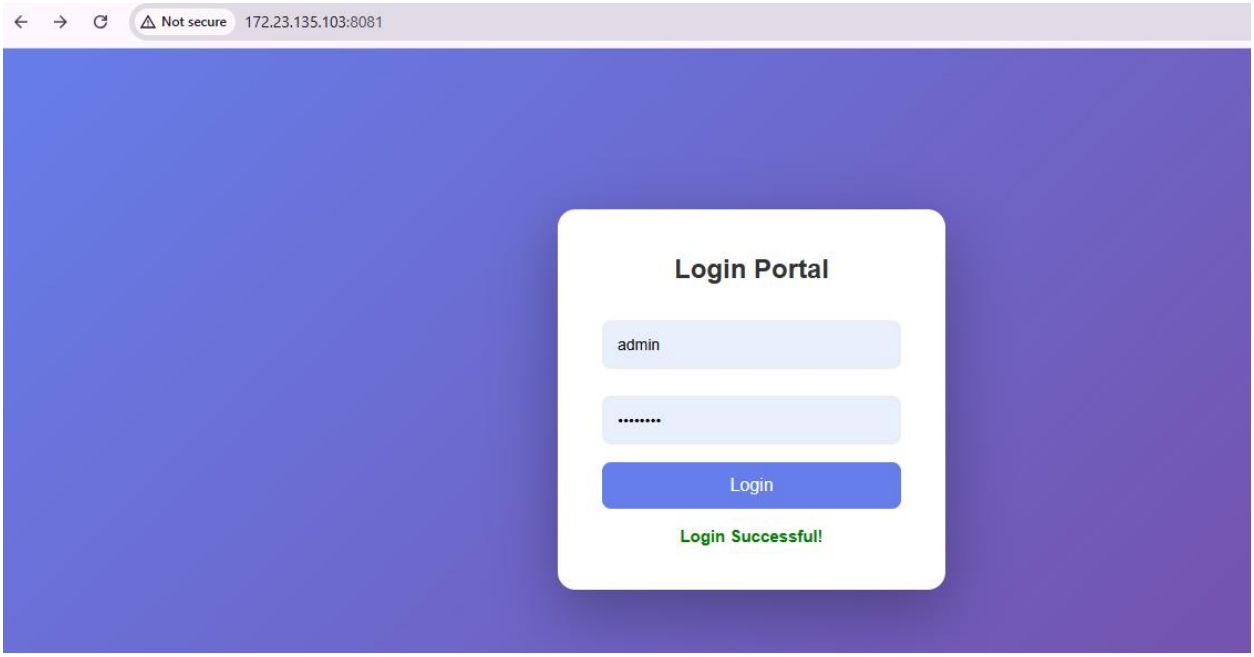
Three tier app have being deployed on Grafana

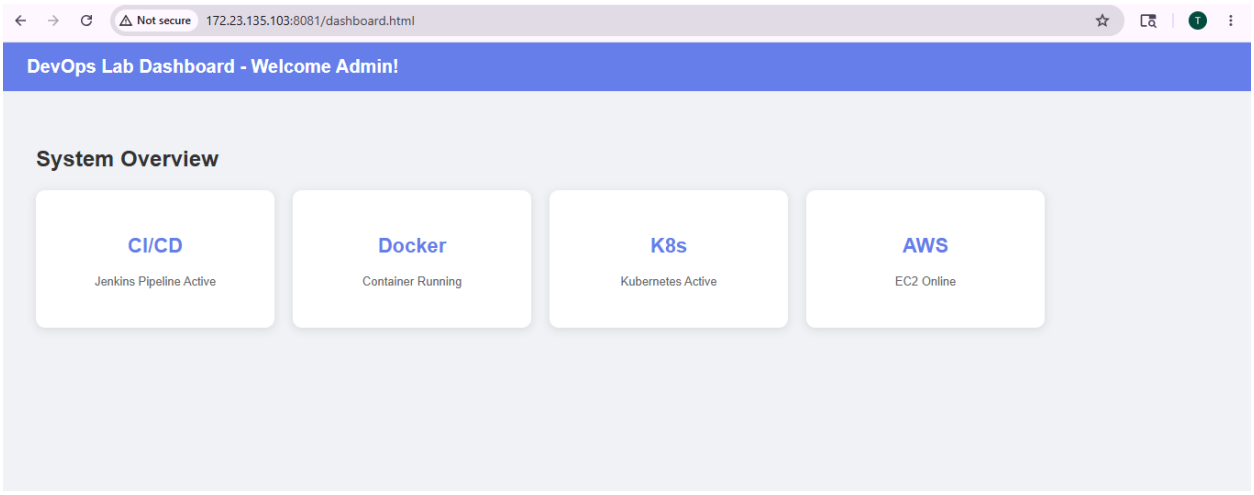


Container metrics



Three tier app run on http://172.23.135.103:8081





Module 8 - DevSecOps Security: Installed Trivy vulnerability scanner and performed security scan on the Docker image to detect HIGH and CRITICAL vulnerabilities.

```
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ wget -qO - https://aquasecurity.github.io/trivy-repo/deb/public.key | sudo a
pt-key add -
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (see apt-key(8)).
OK
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ echo deb https://aquasecurity.github.io/trivy-repo/deb $(lsb_release -sc) ma
in | sudo tee -a /etc/apt/sources.list.d/trivy.list
sudo apt-get update
deb https://aquasecurity.github.io/trivy-repo/deb noble main
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Get:5 https://aquasecurity.github.io/trivy-repo/deb noble InRelease [3235 B]
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:7 https://apt.grafana.com stable InRelease
Get:8 https://aquasecurity.github.io/trivy-repo/deb noble/main amd64 Packages [360 B]
Hit:9 https://apt.releases.hashicorp.com noble InRelease
Fetched 3595 B in 2s (1893 B/s)
Reading package lists... Done
W: https://download.docker.com/linux/ubuntu/dists/noble/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt
/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
W: https://aquasecurity.github.io/trivy-repo/deb/dists/noble/InRelease: Key is stored in legacy trusted.gpg keyring (/et
c/apt/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ sudo apt-get update
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Hit:2 https://aquasecurity.github.io/trivy-repo/deb noble InRelease
```

Activate V

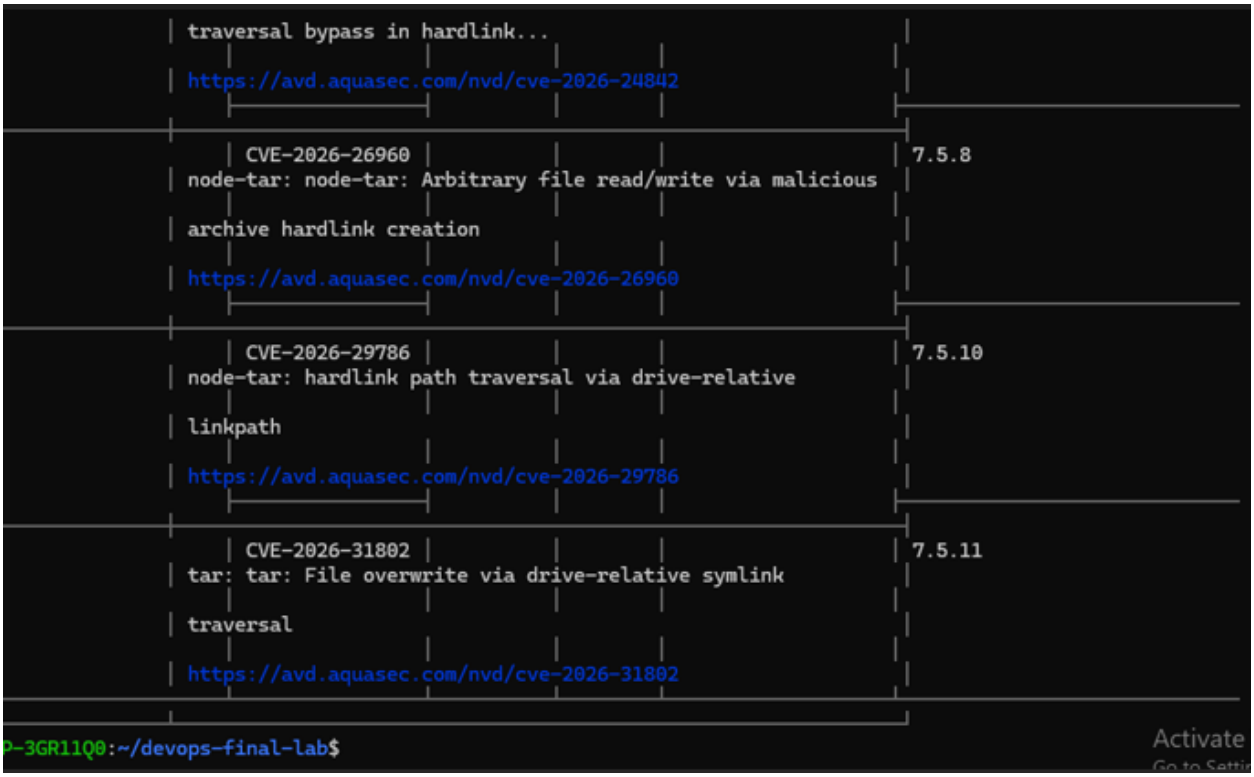
```
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ echo deb https://aquasecurity.github.io/trivy-repo/deb $(lsb_release -sc) ma
in | sudo tee -a /etc/apt/sources.list.d/trivy.list
sudo apt-get update
deb https://aquasecurity.github.io/trivy-repo/deb noble main
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Get:5 https://aquasecurity.github.io/trivy-repo/deb noble InRelease [3235 B]
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:7 https://apt.grafana.com stable InRelease
Get:8 https://aquasecurity.github.io/trivy-repo/deb noble/main amd64 Packages [360 B]
Hit:9 https://apt.releases.hashicorp.com noble InRelease
Fetched 3595 B in 2s (1893 B/s)
Reading package lists... Done
W: https://download.docker.com/linux/ubuntu/dists/noble/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt
/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
W: https://aquasecurity.github.io/trivy-repo/deb/dists/noble/InRelease: Key is stored in legacy trusted.gpg keyring (/et
c/apt/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ sudo apt-get update
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Hit:2 https://aquasecurity.github.io/trivy-repo/deb noble InRelease
Hit:3 https://apt.grafana.com stable InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:6 https://apt.releases.hashicorp.com noble InRelease
Hit:7 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
```

Activate W

```
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ sudo apt-get install trivy -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0 libxcb-d
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  trivy
0 upgraded, 1 newly installed, 0 to remove and 80 not upgraded.
Need to get 48.0 MB of archives.
After this operation, 160 MB of additional disk space will be used.
Get:1 https://aquasecurity.github.io/trivy-repo/deb noble/main amd64 trivy amd64 0.70.0 [48.0 MB]
Fetched 48.0 MB in 1min 3s (762 kB/s)
Selecting previously unselected package trivy.
(Reading database ... 81611 files and directories currently installed.)
Preparing to unpack .../trivy_0.70.0_amd64.deb ...
Unpacking trivy (0.70.0) ...
Setting up trivy (0.70.0) ...
```

```
tehreem@DESKTOP-3GR11Q0:~/devops-final-lab$ trivy image devopslab2026bse/devops-login-app:v1
2026-05-16T19:31:50+05:00       INFO    [vulndb] Need to update DB
2026-05-16T19:31:50+05:00       INFO    [vulndb] Downloading vulnerability DB...
2026-05-16T19:31:50+05:00       INFO    [vulndb] Downloading artifact...        repo="mirror.gcr.io/aquasec/trivy-db:2"
93.06 MiB / 93.06 MiB [-------------------------------------------------------------------] 100.00% 643.69 KiB p/s 2m28s
2026-05-16T19:34:16+05:00       INFO    [vulndb] Artifact successfully downloaded       repo="mirror.gcr.io/aquasec/triv
y-db:2"
2026-05-16T19:34:16+05:00       INFO    [vuln] Vulnerability scanning is enabled
2026-05-16T19:34:16+05:00       INFO    [secret] Secret scanning is enabled
2026-05-16T19:34:16+05:00       INFO    [secret] If your scanning is slow, please try '--scanners vuln' to disable secre
t scanning
2026-05-16T19:34:16+05:00       INFO    [secret] Please see https://trivy.dev/docs/v0.70/guide/scanner/secret#recommenda
tion for faster secret detection
2026-05-16T19:34:23+05:00       INFO    Detected OS     family="alpine" version="3.21.3"
2026-05-16T19:34:23+05:00       INFO    [alpine] Detecting vulnerabilities...   os_version="3.21" repository="3.21" pkg_
num=17
2026-05-16T19:34:23+05:00       INFO    Number of language-specific files       num=1
2026-05-16T19:34:23+05:00       INFO    [node-pkg] Detecting vulnerabilities...
2026-05-16T19:34:23+05:00       WARN    Using severities from other vendors for some vulnerabilities. Read https://trivy
.dev/docs/v0.70/guide/scanner/vulnerability#severity-selection for details.
2026-05-16T19:34:23+05:00       INFO    Table result includes only package filenames. Use '--format json' option to get
the full path to the package file.

Report Summary
```


traversal bypass in hardlink...
https://avd.aquasec.com/nvd/cve-2026-24842
CVE-2026-26960
7.5.8
node-tar: node-tar: Arbitrary file read/write via malicious
archive hardlink creation
https://avd.aquasec.com/nvd/cve-2026-26960
CVE-2026-29786
7.5.10
node-tar: hardlink path traversal via drive-relative
linkpath
https://avd.aquasec.com/nvd/cve-2026-29786
CVE-2026-31802
7.5.11
tar: tar: File overwrite via drive-relative symlink
traversal
https://avd.aquasec.com/nvd/cve-2026-31802
-3GR11Q0:~/devops-final-lab$
Activate